

Kinematic bicycle model:

$$\begin{aligned}\mathbf{x}_k &= [X_k \ Y_k \ v_k \ \psi_k]^\top, \mathbf{u}_k = [a_k \ \delta_k]^\top, \\ X_{k+1} &= X_k + v_k \cos(\psi_k) \Delta t, \\ Y_{k+1} &= Y_k + v_k \sin(\psi_k) \Delta t, \\ \psi_{k+1} &= \psi_k + \frac{v_k}{L} \tan(\delta_k) \Delta t, \\ v_{k+1} &= v_k + a_k \Delta t, \\ \dot{\psi}_k &= \frac{v_k}{L} \tan(\delta_k), \quad \delta_k = \arctan\left(\frac{L \dot{\psi}_k}{v_k}\right).\end{aligned}$$

where X_k and Y_k are the vehicle position coordinates in the global/map frame at time step k ,

v_k is the longitudinal speed of the vehicle,

ψ_k is the yaw angle, or heading angle, of the vehicle in the global/map frame,

a_k is the longitudinal acceleration command,

δ_k is the front wheel steering angle,

$\dot{\psi}_k$ is the yaw rate of the vehicle,

L is the vehicle wheelbase, and in the current MPC code $L = 0.33 \sim \text{m}$,

Δt is the discrete time step, and in the current MPC code $\Delta t = 0.1 \sim \text{s}$.

The constraints used in the current MPC code are:

$$\begin{aligned}-0.4189 &\leq \delta_k \leq 0.4189 \quad [\text{rad}], \\ 0.0 &\leq v_k \leq 6.0 \quad [\text{m/s}], \\ |a_k| &\leq 3.0 \quad [\text{m/s}^2], \\ |\delta_{k+1} - \delta_k| &\leq \pi \Delta t \quad [\text{rad}].\end{aligned}$$